Assignment Report

Task Data and Evaluation

The benchmark is difficult in two different ways. A question can have one correct candidate and three wrong candidates, or all four candidates can agree on the correct final answer. The development data therefore contain both easy unanimous groups and more informative disagreements. The official test split excludes unanimous final-answer groups, which makes its score systematically harder to compare with overall validation accuracy.

Split	Questions	Role in this study
Training	4,755	Fit candidate rankers and learned feature transformations
Validation	798	Compare methods, tune fixed settings, and select the final architecture
Test	467	Score a frozen method; references used only for final scoring

Table 1. Dataset size and permitted use of each split.

Validation is not a clean proxy for test difficulty. It contains 379 questions with one unique final answer, 249 with two answers, 130 with three, and 40 with four. The selected model chose correctly on 100.0%, 95.2%, 80.8%, and 65.0% of those groups, respectively. The test bank removes the easiest category and contains at least two distinct valid answers in every question.

Method

Answer Parsing and Candidate Evidence

I used the supplied parser to extract each candidate's final numeric answer, preserving the official fraction and decimal normalization. Candidates were compared by answer group as well as individually. The feature design combined the fixed 1,024-dimensional whole-trace vector with answer agreement, formatting validity, answer repetition, relative length, formula count, line count, and question number coverage. The parser and scorer were not modified.

Pairwise and Listwise Ranking

The initial verifier was a pairwise linear ranker trained on correct and incorrect candidates from the same question. This optimized relative candidate order. A later listwise objective normalized the four scores jointly and directly assigned probability mass across the four choices. The listwise model improved validation from 92.11% to 92.73% and provided a stronger base for ensemble fusion.

Set Encoder and Selected V2 Ensemble

The second v2 component projects each candidate into a 32-dimensional hidden vector, computes the mean and maximum hidden state across the four candidates, and scores each candidate using its own state plus that question context. It therefore compares a candidate with the other three rather than judging it in isolation. Dropout during training reduced severe overfitting; the final v2 version combines the listwise and set-encoder probabilities with equal weights.

Design choice	Purpose	Inference cost
Linear listwise ranker	Directly optimize four-way selection	One 1,044-dimensional dot product
Four-candidate set encoder	Use pairwise question context and nonlinear interactions	33,537 learned parameters
Equal probability fusion	Reduce variance without ranking-specific learned weights	One weighted average

Table 2. Components of the selected v2 model.

The final decision is the candidate with the highest average probability from the two experts. No question ID lookup, test reference, external model, or online API call is used. The method accepts an arbitrary aligned bank and outputs one integer from zero to three for every question.

Baselines and Ablation

All comparisons below use the same validation split. The baselines establish whether the learned verifier adds information beyond position, length, and answer voting. Vote-longest groups candidates by final answer and breaks a tie by text length. The longest-candidate rule is a surface-style baseline; it does not compare answers across the group.

Baseline	Validation accuracy	Correct / 798	Observation
Random candidate	74.94%	598	Candidate correctness is imbalanced
First candidate	78.20%	624	Position alone is predictive but weak
Longest candidate	77.94%	622	Long explanations do not reliably imply correctness
Vote, tie by longest	86.97%	694	Best non-learned baseline

Table 3. Validation baselines using no reference answers at prediction time.

The answer-voting baseline is useful but leaves 104 validation errors. The fixed candidate verifier was then ablated to separate the contribution of the supplied representation from the explicit structural signals.

Method	Validation	Difference from full
Structure features only	87.84%	-4.26 pp
Embedding only	89.72%	-2.38 pp
Full features	92.11%	Reference
Remove answer consensus	91.60%	-0.50 pp

Table 4. Fixed-seed feature ablation for the first verifier.

The full model outperformed embedding-only by 2.38 percentage points, with a paired bootstrap 95% interval of 0.75 to 4.01 points. This supports combining explicit answer evidence with the fixed representation. Removing length or question-overlap features changed only one validation item, so their contributions were small and should not be overstated.

Official Test Result

The selected v2 checkpoint was frozen before final testing. The official runner produced predictions without opening references, and the independent scorer then checked the parsed final answer. The final score is 380 of 467, or 81.3704%. A fresh reproduction retained every selected candidate except for the runner's variable timing field and again produced the same score.

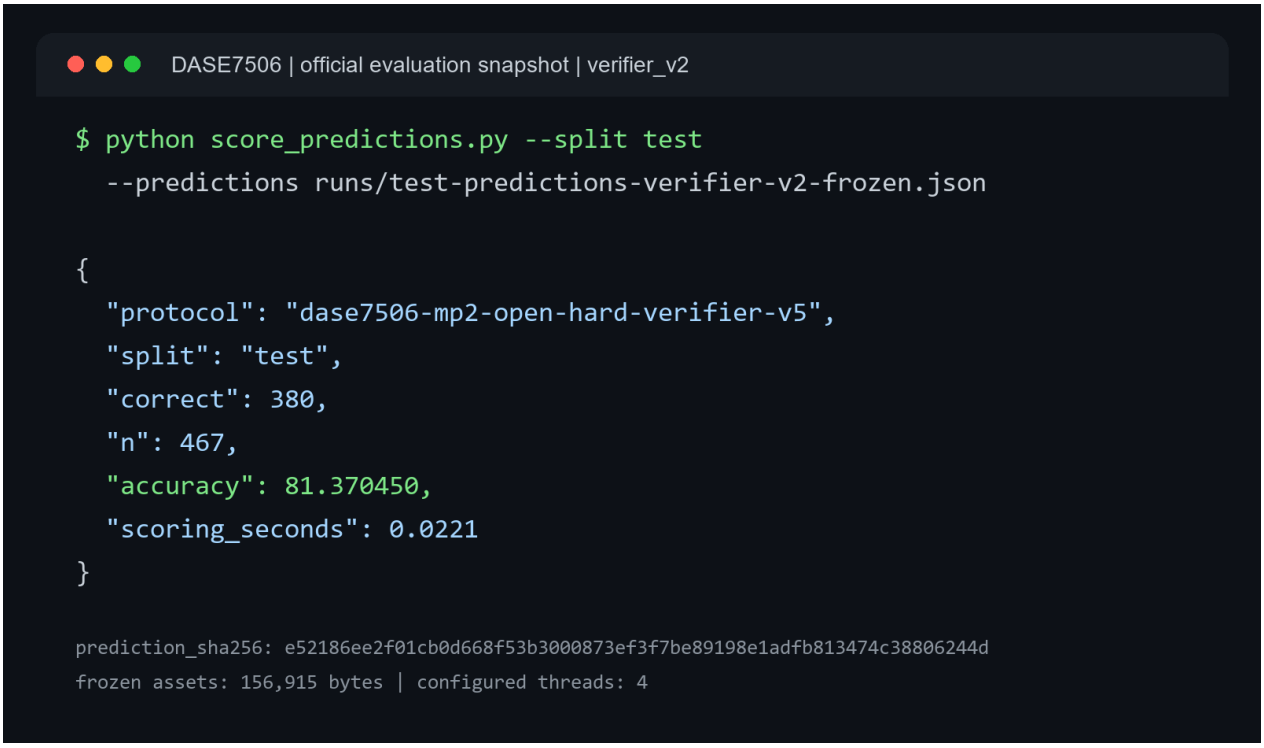


Figure 1. Official scorer output for the selected v2 checkpoint.

The fresh prediction process took 0.64 seconds and scoring took 0.35 seconds, for 0.99 seconds total in the local four-thread run. The v2 checkpoint uses 156,915 uncompressed bytes, far below the 16 MiB asset limit. These measurements include two separate Python process startups.

Selected result	Value
Validation accuracy	747 / 798 (93.61%)
Non-unanimous validation	369 / 419 (87.83%)
Official test accuracy	380 / 467 (81.37%)
Fresh prediction plus scoring	0.99 s

Table 5. Final v2 result summary.

Version Comparison

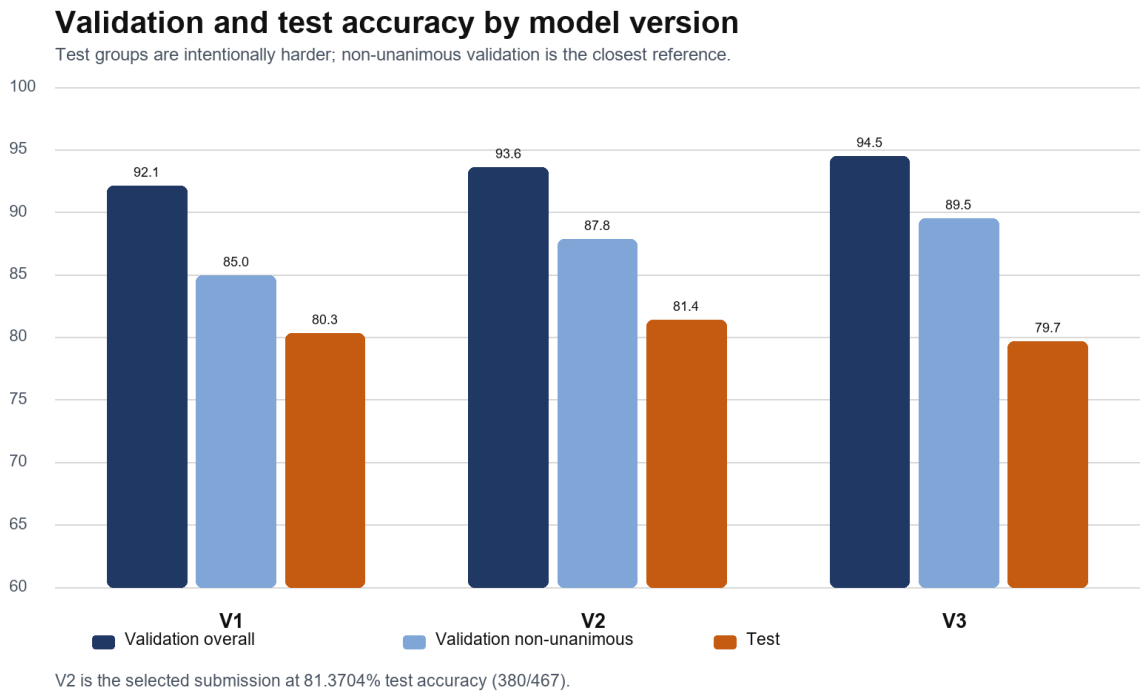


Figure 2. Overall validation, non-unanimous validation, and test accuracy. The middle series is the closest validation reference for the test distribution.

Version	Method	Validation	Test	Test correct
V1	Pairwise candidate ranker	92.11%	80.30%	375 / 467
V2	Listwise + set encoder	93.61%	81.37%	380 / 467
V3	Difficulty-routed experts	94.49%	79.66%	372 / 467

Table 6. Validation and official test accuracy for the three frozen versions.

V2 is the selected submission. V3 produced the highest validation score but lost eight test questions relative to v2. The pattern does not support a claim that more computation causes better selection. It indicates that model capacity was not the limiting resource; transfer across the harder test distribution was.

Why Validation Performance Did Not Transfer

V3 reached 94.49% validation accuracy by routing questions according to the number of distinct final answers. It used v2 for two-answer groups, a two-layer Transformer for three-answer groups, and a group ranker for four-answer groups. This rule was built from small validation subsets. The four-answer branch contained only 40 questions, so a few changes could greatly alter its estimated quality.

Validation group	Questions	V1	V2	V3
All candidates agree	379	100.0%	100.0%	100.0%
Two answers	249	90.8%	95.2%	95.2%
Three answers	130	79.2%	80.8%	84.6%
Four answers	40	67.5%	65.0%	70.0%

Table 7. Accuracy by the number of distinct parsed final answers on validation.

The test score gap cannot be explained by unanimous examples alone. V2 scored 87.83% on the 419 non-unanimous validation questions and 81.37% on test. V3 reached 89.50% on that validation subset but fell to 79.66% on test. The more strongly a method was selected for the observed validation subgroups, the worse its later transfer became.

Error Analysis

Failure category for v2	Errors	Share
Followed a wrong answer with more votes	21	41.2%
Chose the wrong group in a tied vote	19	37.3%
Equal support in non-top groups	6	11.8%
Selected less support than the correct answer	5	9.8%

Table 8. Qualitative failure categories for the 51 validation errors made by v2.

Most errors were confident but systematic. The model often inherited a shared wrong answer from multiple candidate traces. Score margin was not a reliable confidence estimate: the lowest validation probability-margin quintile reached 96.9% accuracy, while the highest reached 90.6%. The score remains useful for ranking candidates within a question, but it should not be presented as a calibrated correctness probability.

Resource Use and Reproducibility

Selected model metric	Measured value	Limit / status
Inference assets	156,915 bytes	Below 16 MiB
Prediction process	0.64 s	Four threads
Official scoring process	0.35 s	Independent process
Combined external time	0.99 s	Below 30 s
Prediction choices reproduced	467 / 467 identical	Same frozen score
Runtime dependency	NumPy 2.5.3	Offline CPU

Table 9. Resource use for the selected v2 model.

The code fixes the model seed at 20261026 and sets OpenMP, OpenBLAS, and MKL to four threads. Validation and test predictions use the official runner. The checkpoint contains the learned linear weights, feature normalization statistics, set-encoder weights, and fusion configuration. Training uses only supplied training inputs and references.

Commands

```
cd code
python -m pip install -r requirements-training.txt
python check_data.py
python experiments/train_ensemble_v2.py
python experiments/train_set_mlp.py
python experiments/build_v2_bundle.py
python predict.py --method submission_v2.py --checkpoint runs/verifier_v2 \
  --split test --output runs/test-predictions-reproduced.json
python score_predictions.py --split test \
  --predictions runs/test-predictions-reproduced.json \
  --output runs/test-score-reproduced.json
```

The supplied parser and scorer remain unchanged. The frozen prediction hash is e52186ee2f01cb0d668f53b3000873ef3f7be89198e1adfb813474c38806244d, and the v2 release manifest records all source and checkpoint hashes.

Limitations and Decision

The main limitation is distribution shift. Test questions are deliberately restricted to answer-disagreement cases, and the validation set is too small to estimate the four-answer branch reliably. A second limitation is that the fixed representation may encode style or question-specific information that does not transfer to harder examples. Arithmetic verification, PCA, gradient boosting, sparse text models, cross-fitted stacking, and larger Transformer models did not produce a validation improvement that justified replacing v2.

I selected v2 because it has the best official score, requires less than one second for complete evaluation, stays far below the asset limit, and reproduces all 467 stored prediction choices. V3 is retained as negative evidence for hard validation routing.

AI Assistance and Reuse

The experiment analysis, and optimization of experimental report were prepared with assistance from AI. All reported scores were generated by the supplied scorer from frozen predictions. The repository contains the source, checkpoints, seeds, hashes, and commands needed for independent reproduction. No external pretrained model, external dataset, or online LLM call was used by the submitted selector.